

Mohammed Alam

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EDUCATION

Columbia University – *Bachelor of Science in Computer Science*

Expected May 2027

Relevant Coursework: Data Structures & Algorithms, Computer Systems, Advanced Programming (C), Database Systems, Web Security, User Interface Design, Software Quality & Design (Java)

EXPERIENCE

Columbia Housing

May 2024 – Present

Senior Systems & Operations

New York, NY

- Digitized operational workflows across 40 buildings and 2M+ sq. ft. via event-driven Node.js services and REST APIs
- Pitched and shipped Columbia's dorm-room viewer for 6,000+ students, winning Housing and CUIT approval to deploy
- Cut manual work 80% across 6,000+ recurring tasks by orchestrating AWS S3, Asana API, and Apps Script pipelines

Headstarter AI

July 2024 – September 2024

Software Engineering Fellow

New York, NY

- Shipped 5 full-stack AI apps in 8 weeks on Next.js, TypeScript, OpenAI, Pinecone, and Stripe payments
- Won 1st of 75+ teams at a TikTok-sponsored hackathon with a real-time React blogging platform built in 48 hours
- Indexed 10,000+ documents for semantic search at <300 ms p95 with an OpenAI-embedding and Pinecone RAG pipeline

Freelance

September 2019 – May 2024

Software Engineer

New York, NY

- Delivered 500+ websites, Discord bots, and automations for 150+ clients using React, Node.js, MongoDB, and Redis
- Lowered API latency 70% across client apps by caching hot MongoDB queries in Redis with a cache-aside pattern
- Shortened mean time to resolution 50% by instrumenting structured logging, error tracking, and alerting

PROJECTS

Dorm Room Viewer | Columbia Housing

June 2025 – Present

- Served 15,000+ room photos and floor plans via an AWS S3 image pipeline with graceful fallbacks
- Enabled Housing staff to self-manage listings for 40 buildings through an admin CMS for photo uploads and room visibility
- Integrated Columbia CAS SSO via NextAuth.js, securing admin routes behind session-based university login

Doorman | Columbia Housing

August 2026

- Launched a TypeScript, React, and Express Asana builder for 40 buildings, now in Housing's daily operations
- Compressed per-building Asana setup from 2 hrs to 1 min by auto-parsing raw room-inventory spreadsheets
- Eliminated manual entry for 6,000+ rooms by auto-detecting buildings, prefixing rooms, and generating floor sections

Mini-Vercel

July 2026

- Architected self-hosted CI/CD turning GitHub pushes into live HTTPS deploys in <60 sec using TypeScript and Docker
- Achieved zero-downtime releases via container health checks and atomic Nginx upstream swaps on AWS EC2
- Routed 10+ project subdomains through a custom Go reverse proxy, managed from a CLI and Next.js dashboard

Guess The Number

October 2021 – June 2024

- Scaled a multiplayer Discord game to 13M+ members across 13K+ servers, processing 1M+ commands/day
- Reduced command latency from 450 ms to 180 ms by sharding Redis and MongoDB state per session to isolate live games
- Maintained sub-200ms p95 response times through 10x traffic spikes by offloading work to Bull.js job queues

SKILLS

Languages: JavaScript, TypeScript, Python, Java, Go, SQL

Frameworks: React, Next.js, React Native, Node.js, Express, NextAuth.js, Bull.js

Data & Infra: PostgreSQL, MongoDB, Redis, Supabase, Pinecone, AWS (EC2, S3), Docker, Nginx, Git, CI/CD, REST APIs